

Technical Project Report: End-to-End Supply Chain Data Pipeline

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Date: July 12, 2026

1. Executive Summary

This project outlines the design and deployment of a modern, cloud-native data pipeline engineered to transform raw supply chain data into high-value business insights. By implementing the Medallion Architecture within Azure Databricks and connecting it to Microsoft Power BI, this solution eliminates data silos.

2. Problem Statement

Before this implementation, supply chain data was fragmented, leading to challenges in inventory management and supplier performance evaluation. The objective was to engineer an automated pipeline to ingest, clean, and visualize data.

3. System Architecture

The solution utilizes the Medallion Architecture:

- **Bronze Layer:** Raw data ingestion.
- **Silver Layer:** Data cleaning, deduplication, and schema enforcement.
- **Gold Layer:** Aggregated, business-ready metrics.

4. Implementation

The pipeline was orchestrated using Azure Databricks with Spark SQL, ensuring scalable computing. Security was managed via Personal Access Tokens (PAT).

5. Business Insights

The Power BI dashboard provides key views: Executive Summary (Revenue Trends), Inventory Status (Stock Alerts), and Supplier Performance (Shipment Volume).

Appendix: SQL Implementation

Silver Layer Transformation

```
CREATE OR REPLACE TABLE silver_supply_chain_data AS
SELECT transaction_id, order_date, product_name,
ABS(quantity) AS quantity
FROM bronze_raw_data;
```

Gold Layer Aggregation

```
CREATE OR REPLACE TABLE gold_sales_trends AS
SELECT DATE_TRUNC('month', order_date) AS month,
SUM(sales_amount) AS revenue
FROM silver_supply_chain_data
GROUP BY 1;
```